

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
COLLEGIUM: A Web Based Marketplace for College Students
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**COLLEGIUM: A Web Based Marketplace for College Students**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Lalit Sharma. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Gurgaon

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Dr. Lalit Sharma (Assistant Professor, Chitkara University)** for his valuable guidance, continuous support, and encouragement throughout the development of this project. His expertise and insights played a crucial role in shaping this work.

We are also thankful to **Chitkara University, Rajpura, Punjab** for providing us with the necessary resources and academic environment to successfully complete this project.

We would like to acknowledge the efforts of all team members:

- Sneha Bhatia (2210992389)
- Ankur Samir (2210991281)
- Tushita (2210990915)

Their collaboration, dedication, and teamwork were essential in completing this project successfully.

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Abstract

Collegium is a web-based marketplace developed exclusively for college students, designed to create a secure and convenient platform where students can buy, sell, and exchange products and services within their academic community. The platform allows students to list items such as books, electronics, notes, stationery, furniture, and other useful college-related products that are no longer needed, making them accessible to other students at affordable prices.

Unlike general online marketplaces, Collegium focuses specifically on student needs by building a trusted campus-oriented ecosystem where transactions are carried out among verified users. This not only increases reliability and safety but also encourages a sense of community and collaboration among students.

The platform aims to solve common student challenges such as limited budgets, difficulty in finding affordable study resources, and lack of a dedicated space to exchange second-hand goods. By promoting reuse and resale, Collegium also supports sustainable consumption and reduces unnecessary waste.

1. Introduction

College students frequently require affordable products such as textbooks, study materials, electronics, stationery, furniture, and daily-use items, but purchasing these products from traditional retail stores or large online marketplaces can often be expensive and inconvenient. At the same time, many students possess items that are still in good condition but are no longer useful to them, such as old textbooks, lab equipment, hostel essentials, gadgets, or classroom notes. These items often remain unused or are discarded, despite being valuable to other students.

Existing online marketplaces provide a broad platform for buying and selling goods, but they are generally designed for a wide audience and do not specifically address the unique requirements of college students. Such platforms often lack student verification, campus-based filtering, affordable pricing expectations, and a community-focused environment, making transactions less relevant and sometimes less trustworthy for students.

Collegium addresses this gap by introducing a dedicated student marketplace ecosystem that is designed specifically for college communities. It provides a focused platform where students can connect with fellow students to buy, sell, and exchange products in a secure and reliable environment. By creating a campus-oriented digital marketplace, Collegium promotes affordability, accessibility, convenience, and peer-to-peer trust, while also encouraging the reuse of products and fostering a collaborative student network.

2. Methodology

The project follows a modular development approach:

1. **Requirement Analysis:** Identifying student needs for a secure and affordable campus marketplace.
2. **System Design:** Designing platform architecture, database, and user interface modules.
3. **Development:** Building features such as user login, product listing, search, and communication modules.

4. **Integration:** Combining all modules into one unified marketplace platform. The system uses computer vision techniques to detect exercises and track user activity in real time.
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3. Tools and Technologies

- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend:** Node.js / Java
- **Computer Vision:** Python, OpenCV
- **Database:** MongoDB / SQL
- **Version Control:** Git, GitHub
- **Deployment:** Cloud platforms

GitHub Repository:

<https://github.com/ankur1205/Collegium-2210991281-2210990915-2210992389-.git>

4. Implementation

The system is implemented as a full-stack web application consisting of multiple modules:

- **User Authentication Module:** Secure student login and registration
- **Product Listing Module:** Uploading and managing products for sale or exchange
- **Search & Filter Module:** Easy product discovery through categories and search options
- **Communication Module:** Direct interaction between buyers and sellers
- **Order / Deal Management:** Managing product availability and transaction status
- **User Profile Management:** Personalized user details and activity tracking

Each module is integrated to provide a smooth, secure, and user-friendly marketplace experience for college students.

5. Major Findings / Outcomes / Results

Successfully developed a working web-based marketplace platform

- Enabled easy buying, selling, and exchange of products among students

- Reduced student expenses through affordable second-hand products
- Promoted reuse and sustainable utilization of useful items
- Built a trusted campus-focused marketplace community

The system demonstrated effectiveness in creating a convenient, affordable, and student-friendly digital marketplace.

6. Conclusion

Collegium successfully delivers a dedicated digital marketplace platform for college students. It provides students with a secure and convenient environment to buy, sell, and exchange products while promoting affordability, trust, and a collaborative campus community.

7. Future Scope

- Mobile application development for easier access and convenience
 - Integration of secure payment gateway for smooth transactions
 - AI-based product recommendations for personalized user experience
 - Campus verification system to enhance trust and security
 - Multilingual support for wider accessibility among students
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8. References

- React.js Official Documentation
 - Node.js Documentation
 - MongoDB Documentation
 - Research papers on E-Commerce and Online Marketplace Systems
 - GitHub Repository: Collegium Project Source Code
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